Problem 18.20

Jim buys a perpetuity of 100 per year, with the first payment 1 year from now. The price for the perpetuity is 975.61, based on a nominal yield of i compounded semiannually. Immediately after the second payment is received, the perpetuity is sold for 1642.04, to earn for the buyer a nominal yield of j compounded semiannually. Calculate $i - j$.

Solution

The first purchase is a level perpetuity-immediate with annual payments of 100. Let $i^{(1)}$ be the annual effective rate corresponding to the nominal rate i compounded semiannually. Then

$$975.61 = 100a_{\overline{\infty}|} = 100\left(\frac{1}{i^{(1)}}\right).$$

Thus,

$$i^{(1)} = \frac{100}{975.61} = 0.1024999744.$$

Since i is nominal compounded semiannually,

$$1 + i^{(1)} = \left(1 + \frac{i}{2}\right)^2.$$

Therefore,

$$i = 2\left(\sqrt{1.1024999744} - 1\right) \approx 0.100000.$$

So

$$i \approx 10\%.$$

After the second payment is received, the perpetuity is sold. From the buyer's point of view, the next payment is 1 year later, so the sale price is again the present value of a perpetuity-immediate. Let $j^{(1)}$ be the annual effective rate corresponding to the nominal rate j compounded semiannually. Then

$$1642.04 = 100a_{\overline{\infty}|} = 100\left(\frac{1}{j^{(1)}}\right).$$

Thus,

$$j^{(1)} = \frac{100}{1642.04} = 0.0608998563.$$

Since j is nominal compounded semiannually,

$$1 + j^{(1)} = \left(1 + \frac{j}{2}\right)^2.$$

Therefore,

$$j = 2\left(\sqrt{1.0608998563} - 1\right) \approx 0.060000.$$

So

$$j \approx 6\%.$$

Hence,

$$i - j \approx 0.10 - 0.06 = 0.04.$$

Therefore,

$$\boxed{4\%}.$$